

Prabal Ghosh

Nice, France

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Research Interests

Multi-modal imaging and Signal Processing, Computer Vision & vision-language models, Data Science, NLP, Semantic Web

Education

- **MSc in Data Science and Artificial Intelligence (M2)** Sept 2024 – Sept 2025
Université Côte d'Azur, France
Final Grade: 14.97 / 20
Thesis: *MRI Analysis for Disease Progression Assessment in FSHD*.
courses: *Bayesian Learning, Reinforcement Learning, Biomedical Signal Processing, Model Based Statistical Learning, Application of ML to MRI electroencephalography and Brain Computer Interfaces, Inverse Problems in Images, Medical Images, ML for Multi-omics, Topological data analysis*.
- **MSc in Data Science and Artificial Intelligence (M1)** Sept 2023 – Aug 2024
Université Côte d'Azur, France
Final Grade: 12.95 / 20
courses: *Algebra, Probability, Statistical Inference, Machine learning, R, Ethical of data, Big Data Systems, Information Theory, Model selection and resampling, Deep Learning, Optimization*.
- **Master of Computer Applications (MCA)** Sept 2020 – Aug 2022
Amrita Vishwa Vidyapeetham, India
Final Grade: 95%
Thesis: *Mathematical Modeling of Parkinson's Disease*.
courses: *Python, Java, Operating Systems, Computer Networks, DataBase, Data Structures and Algorithms, Data Mining, Internet of Things, Natural Language Processing, Semantic Web*.
- **Bachelor of Technology in Electronics and Communication Engineering** Aug 2014 – Oct 2018
RCC Institute of Information Technology, India
Final Grade: 78.60%
courses: *Analog and Digital Signal Processing, Analog & Digital Electronics, Analog & Digital Communication, Control System, Wireless Communication, EM Theory, Microwave Engineering, Radar Systems*.

Honors and Awards

- **University Gold Medal** (Ranked 1st class 1st in the University), Master of Computer Applications (MCA), Amrita Vishwa Vidyapeetham, India.
- **IDEX (Initiative d'Excellence) Excellence Scholarship**, Université Côte d'Azur, France, awarded for academic excellence during the Master's program.

Research Experience

INRIA – Sophia Antipolis, France

Apr 2025 – Sept 2025

MRI Analysis for Disease Progression Assessment in FSHD

Team- EPIONE

Supervisor- Marco Lorenzi

- Developed a harmonized MRI preprocessing pipeline for Facioscapulohumeral Muscular Dystrophy (FSHD), integrating bias correction, intensity harmonization, segmentation, and quantitative biomarker extraction.

- Investigated the interpretability and robustness of concept-based explainability methods for CNN.

Industrial Experience

Freshworks Inc., India

Apr 2022 – Mar 2023

Software Engineer

- Worked on scalable system using Python Django and Ruby on Rails.

Publications

- Ghosh, P., Sasidharan, R. S., Rohit, P. S., Nair, M. (2023). *A study on modeling the dopaminergic pathways of Parkinson's disease.*

Technical Skills

- **Programming:** Ruby, Python, Java, R, MATLAB, JavaScript, HTML, CSS
- **Databases:** PostgreSQL, MySQL, MongoDB, SQL, PL/SQL, RDFS, OWL, SPARQL, SHACL
- **Frameworks & Tools:** Django REST Framework, Ruby on Rails, 3D Slicer
- **Libraries:** PyTorch, TensorFlow, Keras, scikit-learn, NLTK, SciPy, NiBabel, SimpleITK, Nilearn
- **Domains:** MRI Analysis, EEG/ECG Signal Processing, NLP, Computer Vision, Medical Imaging

Academic Projects

- Machine Learning for Climate Change: ML-based analysis of building energy efficiency and climate impact.
- Semantic Modeling using OWL: Ontology-based semantic modeling for a recipe box company.
- Pet Finder using ML & DL: Deep learning-based pet classification and prediction system.
- Autonomous Taxi Agent: Reinforcement learning-based autonomous agent.
- Physically-Informed Machine Learning: Modeling plant–pathogen dynamics using omics data.
- Semantic web project: Linked data extraction from Springer using OWL and SPARQL

Certifications & Training

- Deep Learning Summer School: Université Côte d'Azur, SophiaTech
- Machine Learning Specialization: University of Washington (Coursera)
- Mathematics for Machine Learning: Imperial College London (Coursera)
- Python Data Structures: University of Michigan (Coursera)
- Neural Networks and Deep Learning: DeepLearning.AI (Coursera)
- Introduction to a Web of Linked Data: France Université Numérique (INRIA)

Languages

English (Fluent) — French (A1) — Hindi (Fluent) — Bengali (Native)

References

Marco Lorenzi

Research Scientist(CR)

INRIA – EPIONE Research Team

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Remya Rajesh

Assistant Professor (Senior Grade)

School of Computing

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